CT Configuration — Reference Alignment Note

Project: NITK-CTSim | Prepared by: CT Configuration Team | Date: 28 September 2026

Purpose. This note shows how the simulated system configuration relates to the two real reference scanners supplied for this project. The configuration is a simulation-oriented design; it is not a reproduction of either scanner.

Reference basis. GE HealthCare Revolution CT / Apex is the primary architectural reference (single-source, like our simulation). Siemens SOMATOM Force is dual-source, so it is used as a range reference only.

Parameter	Our value	GE Revolution CT	Siemens SOMATOM Force	Basis
Rotation time	0.28 s	0.28 s / 0.23 s (Apex)	0.25 s (minimum)	Matched (GE)
Tube voltage	120 kVp	70/80/100/120/140 kV	70–150 kV	Matched (GE step; Siemens range)
Source-to-detector (SDD)	949 mm	~950 mm	976 mm	Matched (GE); Siemens −2.8%
Source-to-isocenter (SID)	541 mm	~535 mm	535 mm	Close (+1.1% vs both)
Display FOV	500 mm	Not specified	500 mm	Matched (Siemens)
Tube current	200 mA (60–660)	10–1300 mA	20–1300 mA per tube	Within reference range
Pitch	0.516	0.15–1.531	0.15–3.2	Within reference range
Scan range	350 mm	Up to 2000 mm	Up to 2000 mm	Within reference range
Bore diameter	820 mm	800 mm	780 mm	Team-selected (slightly larger)
Collimation	40 mm	160 mm z-coverage	57.6 mm z-coverage (dual)	Team-selected
Views per rotation	984	Not published	Not published	Team-selected
Exposure time / total mAs	1.0 s / 200 mAs	Not published	Not published	Team-selected
Tube current modulation	On; depth 25%; phase 0°	Not published	Not published	Team-selected

Green rows: values matched to the reference systems. Unshaded rows: within reference range or selected by the team where manufacturers do not publish the value.

Derived helical parameters

- Geometric magnification (SDD/SID): 1.754 (GE 1.776; Siemens 1.824).
- Angular increment 0.366°; table feed 20.64 mm per rotation; table speed 73.71 mm/s.
- 17 rotations, 16,728 total views, 4.76 s total scan time for the 350 mm scan range.

Notes on team-selected values

- Collimation of 40 mm equals the Detector module’s active height (64 rows × 0.625 mm), keeping the two modules consistent. It is smaller than both references’ z-coverage.
- The nominal exposure setting (1.0 s, 200 mAs) is a protocol value separate from the derived helical scan duration of 4.76 s.
- Bore diameter (820 mm) is slightly larger than both references (800 mm and 780 mm).
- View count and tube current modulation settings have no published equivalent in either datasheet.

Sources: Siemens SOMATOM Force and GE HealthCare Revolution CT / Apex technical reference datasheets; CT Configuration module defaults (ct_configuration.py).